NAUFALDO, M.Sc.

Cold Storage & Automation Engineer • Asesor LSP TPTU • Robotics Researcher

Bandung, Indonesia • naufaldo1301@gmail.com • dinginlestariteknik.com • linkedin.com/in/naufaldo-do-80b39b1a5 • github.com/Naufaldo

EXECUTIVE SUMMARY

Mechatronics & Automation Engineer (M.Sc. NTUT Taiwan) and Managing Director at CV. Dingin Lestari Teknik. Official HVAC & Refrigeration Competency Assessor (PT. LSP TPTU, Reg. MET.000.002933 2026), F-Gas Level 4 Technical Specialist, and Registered Software Patent Holder for MATLAB Direct Air Capture (DAC) telemetry. Over 7 years of industrial experience delivering 15+ commercial & industrial refrigeration plants (Freezer Rooms, Chiller Rooms, ABF), 3-phase electrical automation panels, enterprise field-service ERP software, and 7 peer-reviewed international robotics publications.

CORE TECHNICAL COMPETENCIES

Refrigeration & Thermal: Bitzer/Copeland compressor sizing, load calculation, Chiller & Freezer Rooms (-25°C), ABF (-40°C), F-Gas L4.

Robotics & Control Systems: Multi-agent formation, Swarm UAVs, APF obstacle avoidance, LiDAR SLAM, ROS, OpenCV.

Electrical & Controls: 3-phase industrial panel wiring, VFD, Modbus RTU/TCP, PLC, safety interlocks, IoT telemetry.

Software & Tools: MATLAB/Simulink (Patent), Python, C/C++, TypeScript, Web ERP Architecture, AutoCAD Electrical.

PROFESSIONAL EXPERIENCE

Managing Director — CV. Dingin Lestari Teknik *June 2025 – Present | Bandung, ID*

- Direct engineering design, thermodynamic load calculations, 3-phase power control panel manufacturing, and Bitzer/Copeland compressor commissioning for industrial Freezer Rooms, Chiller Rooms, and Air Blast Freezers (ABF).
- Architected DLT ERP (demo.dinginlestariteknik.com), a web platform for cold storage maintenance dispatch and inventory.

Official Competency Assessor — PT. LSP TPTU *2026 – Present | Indonesia*

- Officially certified competency assessor for HVAC & Industrial Refrigeration (Registration No. MET.000.002933 2026).
- Evaluate engineering competency and national certification assessments under BNSP standards.

Industrial Electrician & Controls Engineer — CV. Dingin Lestari Teknik *Nov 2018 – June 2025 | Bandung, ID*

- Fabricated and wired 50+ 3-phase electrical control panels, circuit breaker coordination, digital temperature controllers, and VFDs.
- Diagnosed and serviced commercial refrigeration compressors, piping refrigerant circuits, and emergency shutdowns.

Co-founder & Hardware Lead — Beantrack *Sept 2023 – Nov 2024 | Indonesia*

- Designed solar-powered smart agricultural IoT hardware for precision temperature & humidity regulation in coffee dryers.

Cloud Computing Fellow — Bangkit Academy (Google, GoTo, Traveloka) *Feb 2023 – July 2023 | Indonesia*

- Completed Cloud Computing specialization with Distinction; capstone project recognized in Top 20 nationally with business incubation.

EDUCATION

National Taipei University of Technology (NTUT) — Taipei, Taiwan *2024 – 2025*

Master of Science (M.Sc.) in Mechatronics Engineering • GPA: 3.9+/4.0 • Focus: Multi-Agent Robotics & Swarm Control

Politeknik Manufaktur Bandung (POLMAN) — Bandung, Indonesia *2019 – 2023*

Sarjana Terapan Teknik (S.Tr.T) in Automation & Manufacturing Systems

PATENTS & PROFESSIONAL CERTIFICATIONS

- **Registered Software Patent:** MATLAB Telemetry & Direct Air Capture (DAC) Biochar Control Software (Kemenkumham RI).
- **Official Competency Assessor Certificate:** PT. LSP TPTU / BNSP (No. Reg. MET.000.002933 2026).
- **Technical Refrigeration Certification:** Teknik Refrigerasi F-Gas Tingkat 4 (Advanced Level 4).

SELECTED PEER-REVIEWED SCIENTIFIC PUBLICATIONS

- **[IJCAS 2026]** *Robot Coordination in Multi-agent Systems: Leader-Dependent Following Configuration Control of Mecanum Wheeled Drive Robots*, Springer Nature (DOI: 10.1007/s12555-026-00119-1).
- **[SICE FES 2025]** *Dynamic Goal-Shifting and Sinusoidal Altitude Modulation for Flight Control of a Swarm of Quadcopters with Flocking Dynamics*, IEEE (DOI: 10.23919/SICEFES67750.2025.11236621).
- **[ICCAS 2025]** *Fuzzy-Tuned PID Control for Dynamic Formation of Differential-Wheeled Mobile Robots with APF Obstacle Avoidance*, IEEE (DOI: 10.23919/ICCAS66577.2025.11301156).
- **[JCIE 2025]** *Leader-follower Configuration Based Formation Control of Multi-Agent Systems*, Taylor & Francis (DOI: 10.1080/02533839.2025.2503867).
- **[ICCAS 2024]** *Comparative Analysis of Autonomous Indoor Exploration Strategies: Floodfill vs. Frontier-Based Method*, IEEE (DOI: 10.23919/ICCAS63016.2024.10773212).